

Research Proposal: AI-Driven Temporal Modeling of Immune-Cancer Cell Interactions

1. Background and Motivation

Antibody-based cancer therapies have demonstrated immense potential, yet patient response remains highly variable. Traditional static microscopy provides only snapshots of the immune response, failing to capture the complex, dynamic spatial-temporal behaviors of immune cells as they attack malignant cells. To understand why immune cells succeed or fail, we must transition from static image analysis to continuous-time video modeling. My background in developing continuous-time State-Space Models (Medical_MambaTS) and Spatial-Temporal Transformers for complex digital twins positions me to build the computational architecture required to decode these live-cell microscopy videos.

2. Research Objectives

This project aims to develop an Explainable AI (XAI) pipeline to track, model, and interpret live-cell interactions between immune and cancer cells. The specific objectives are: 1. **Spatial-Temporal Cell Tracking:** Develop robust computer vision algorithms using deep learning to accurately segment and track individual immune and cancer cells across time-lapse microscopy videos, even in dense or occluded environments. 2. **Behavioral Sequence Modeling:** Apply continuous-time sequence models (such as Mamba/State-Space Models or Transformers) to the extracted trajectories to classify the behavioral phenotypes of immune cells during an attack (e.g., successful engagement, evasion, or exhaustion). 3. **Explainable Decision Making:** Implement XAI techniques to highlight the specific spatial cues, temporal dynamics, and interaction patterns that correlate with successful cancer cell elimination, providing actionable insights for experimental biologists.

3. Proposed Methodology

- **Data Processing & Tracking:** Utilizing modern frameworks like PyTorch and MONAI, I will implement instance segmentation architectures (e.g., Mask R-CNN or YOLO-based trackers) adapted for phase-contrast or fluorescence microscopy.
- **Temporal Dynamics Extraction:** To handle the temporal dimension, I will extract spatial coordinates and morphological features of the cells over time. Instead of relying solely on standard RNNs, I propose using State-Space Models (SSMs) to model the continuous-time dynamics of the cell interactions, allowing the network to capture long-term dependencies in the immune response without quadratic memory scaling.
- **Explainability:** To ensure the models are useful for biologists, I will integrate attention mapping and feature attribution methods (e.g., Integrated Gradients) to visualize *why* the AI predicts a certain interaction outcome.

4. Expected Impact

By applying advanced machine learning and temporal sequence modeling to live-cell videos, this research will uncover the hidden behavioral rules governing immune-cancer interactions. These insights will directly support biologists in refining antibody treatments, ultimately contributing to more reliable and personalized cancer immunotherapies.